Provision Virtual Machine & Attach Storage

Platform Used: Microsoft Azure (Sandbox Environment)

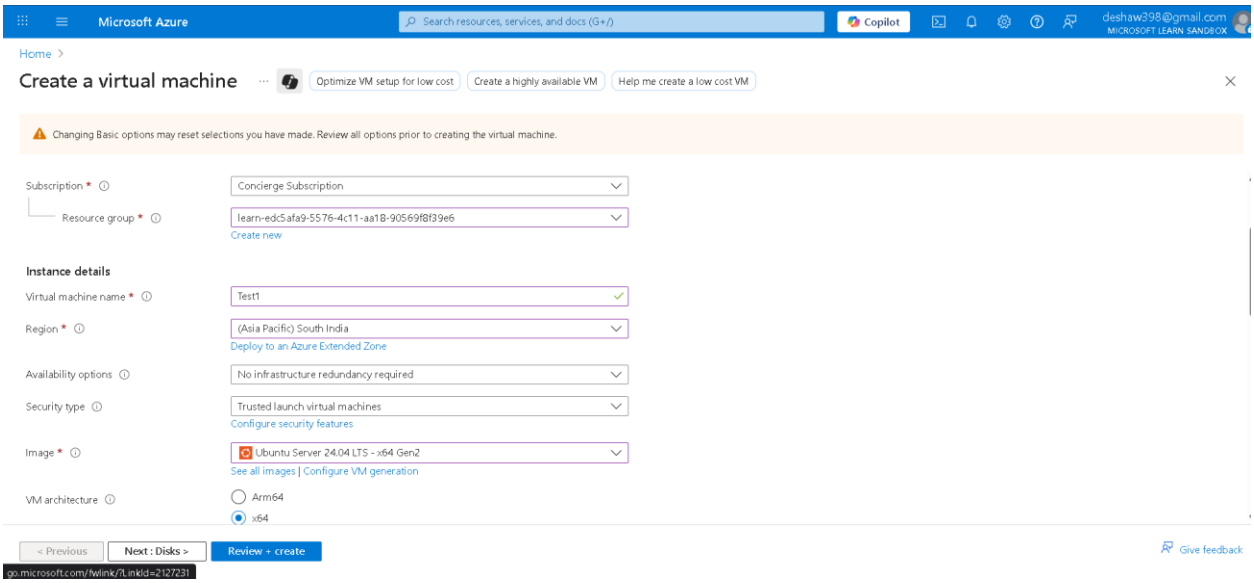
Objective

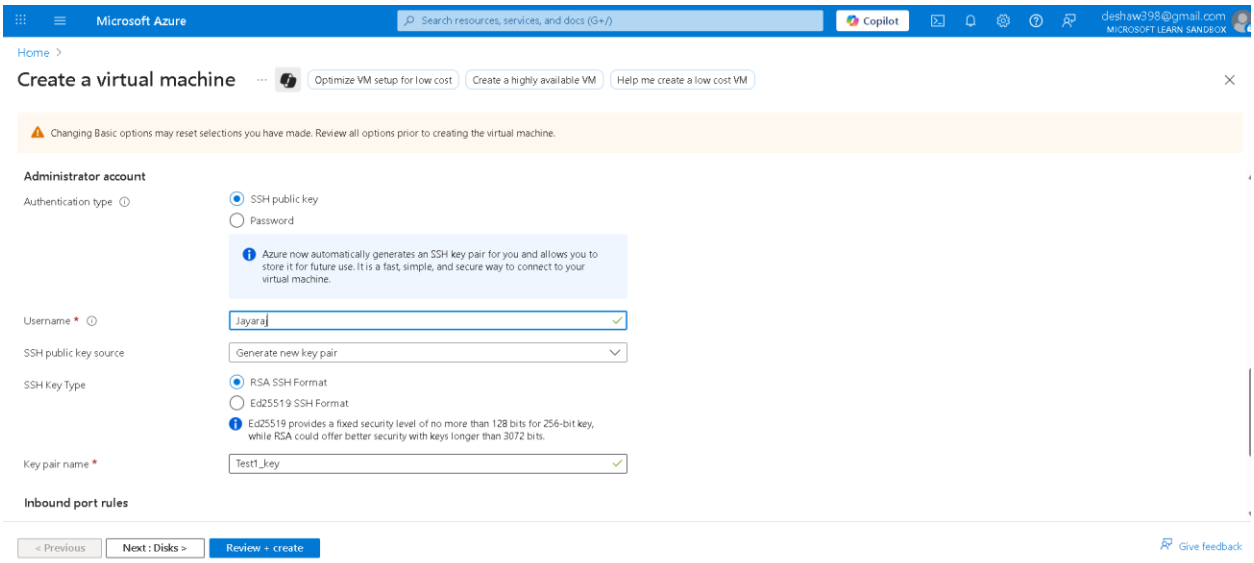
To provision a Virtual Machine (VM) in Azure, securely connect to it via SSH, and expand its capabilities by attaching additional block storage.

Sub-Tasks

1. Selecting OS

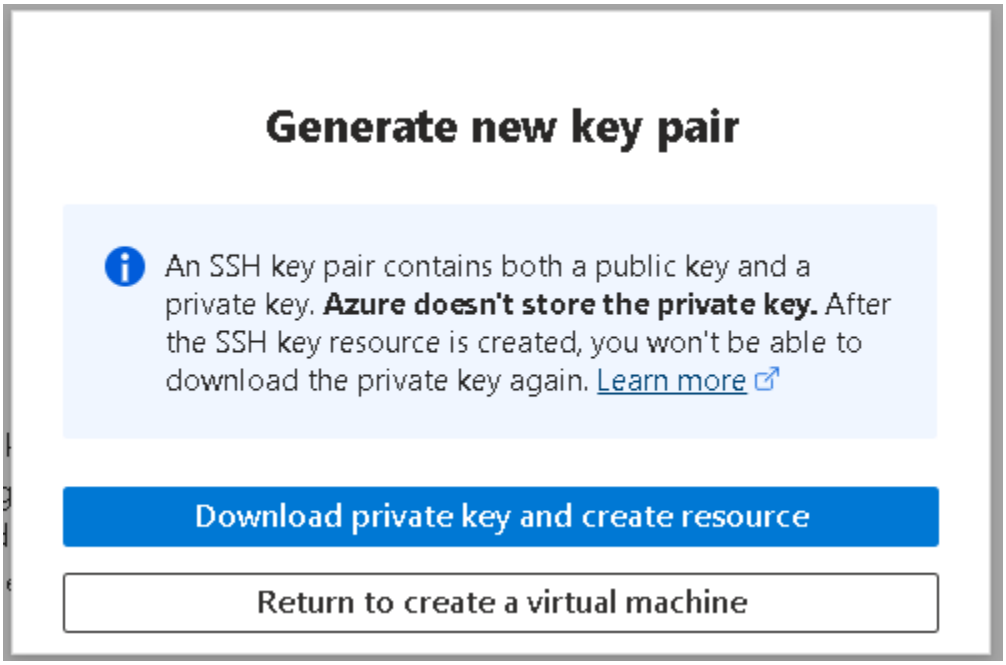
- Action: Selected Ubuntu Server 20.04 LTS as the base operating system.
- Reason: Ubuntu is a widely supported, stable, and secure Linux distribution ideal for development, DevOps, and cloud workloads.
- Steps Taken:
    - In the Azure portal, navigated to Azure services > Virtual Machine.
    - Insert credentials such as subscription, name of machine, user, ssh,etc
    - Choose Ubuntu 20.04 LTS from the list of available images.





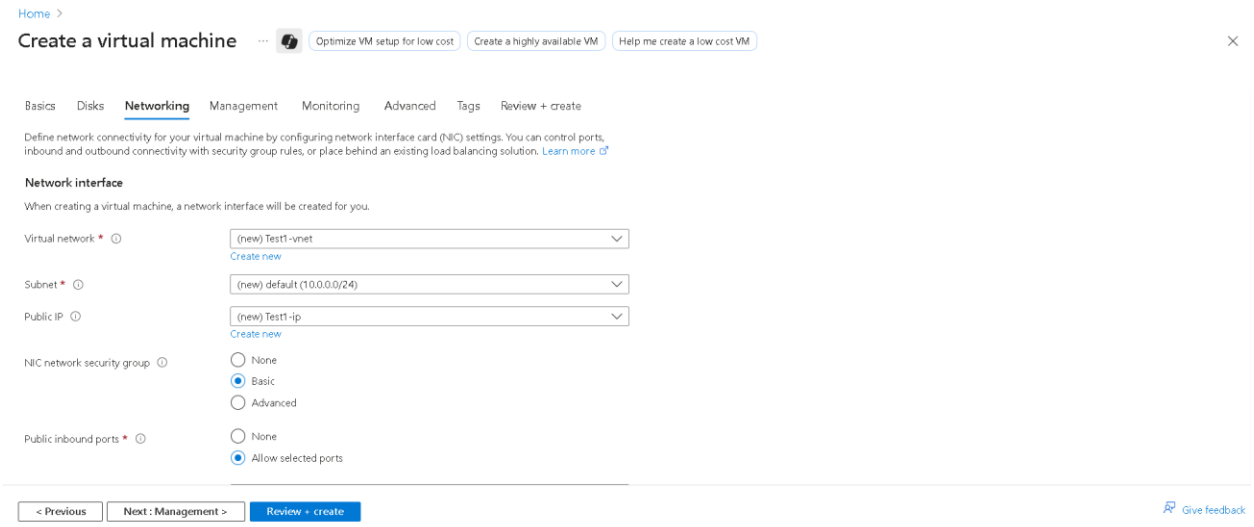
2. Create/Select Key Pair for SSH

- **Action**: Created a new SSH key pair for secure remote access.
- **Reason**: SSH keys offer a more secure method than password authentication.
- **Steps Taken**:
  - Opted for **SSH public key** authentication during VM creation.
  - Downloaded the private key `.pem` file securely and saved it for use with an SSH client.



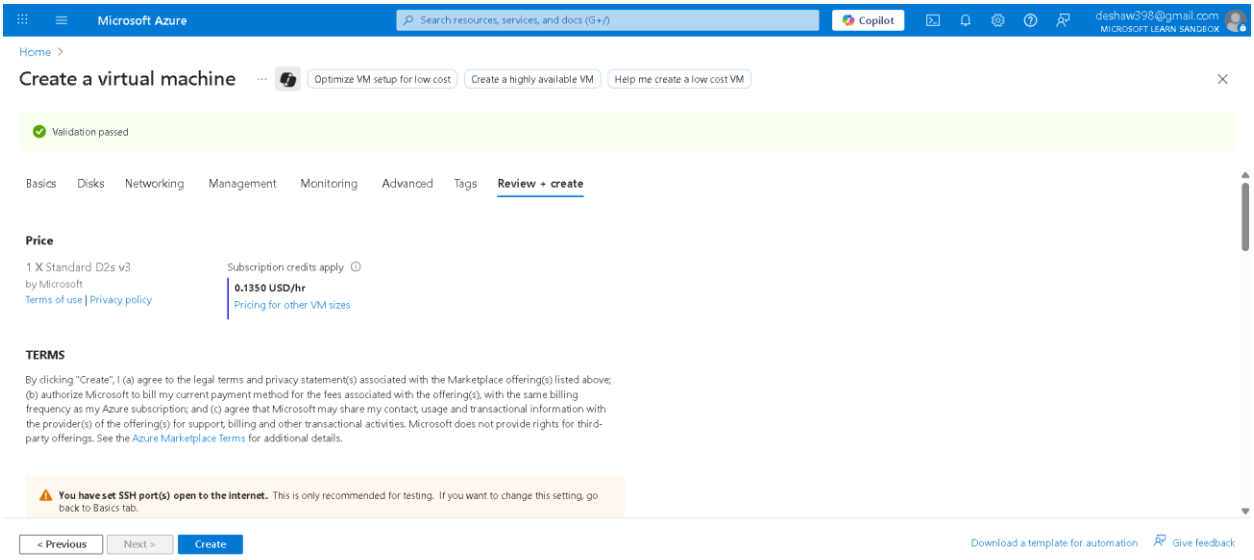
3. Configure Network & Security Group

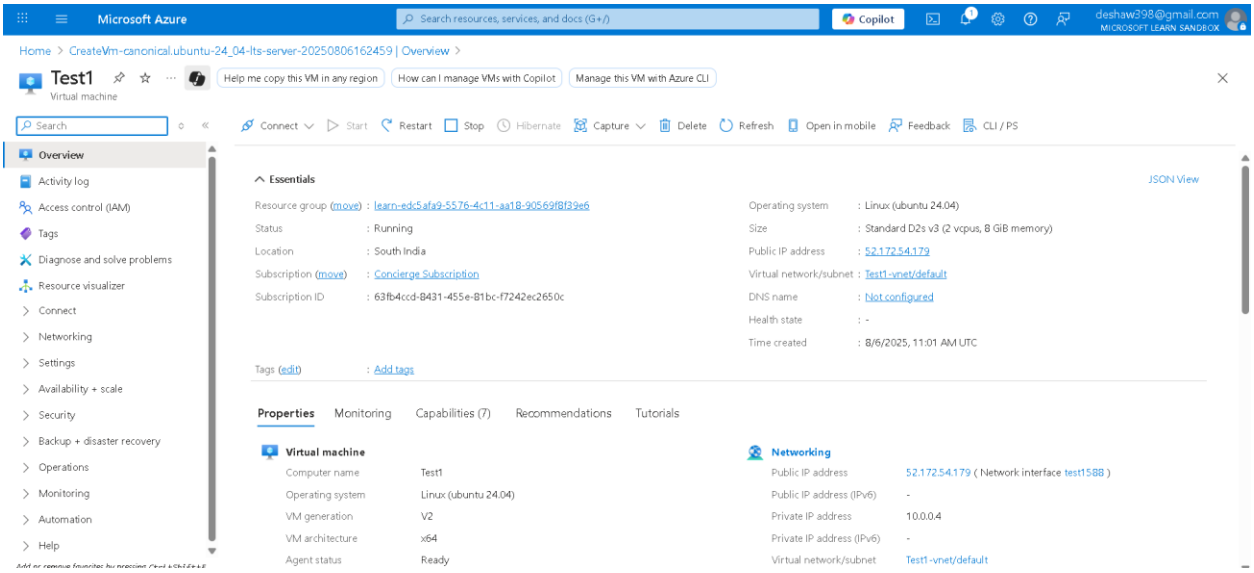
- **Action**: Config network to access the VM via ssh
- **Reason**: To access remote access to VM
- **Steps Taken**:
    - select : virtual network, subnet, public ip, NIC NSG and public inbound port



4. Launch VM & Get Public IP

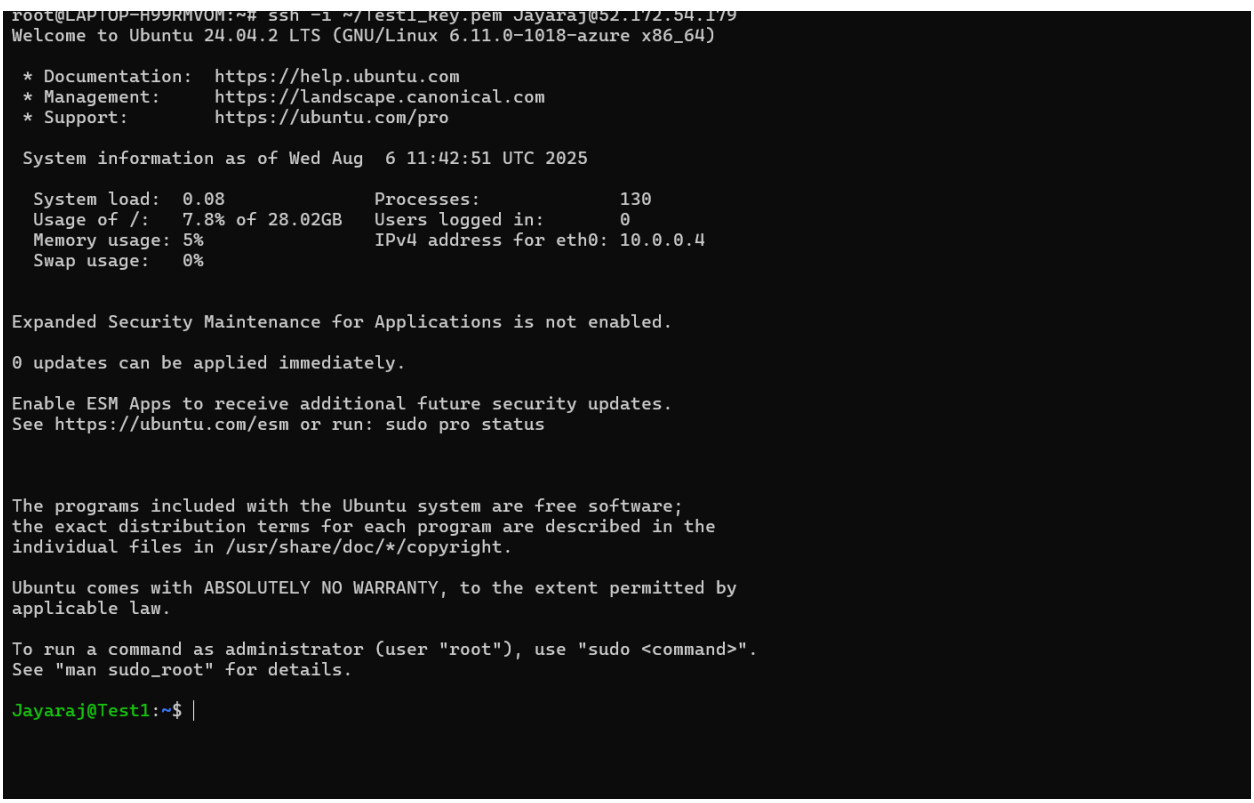
- **Action**: Successfully launched the virtual machine.
- **Output**: Azure assigned a public IP address to the VM.

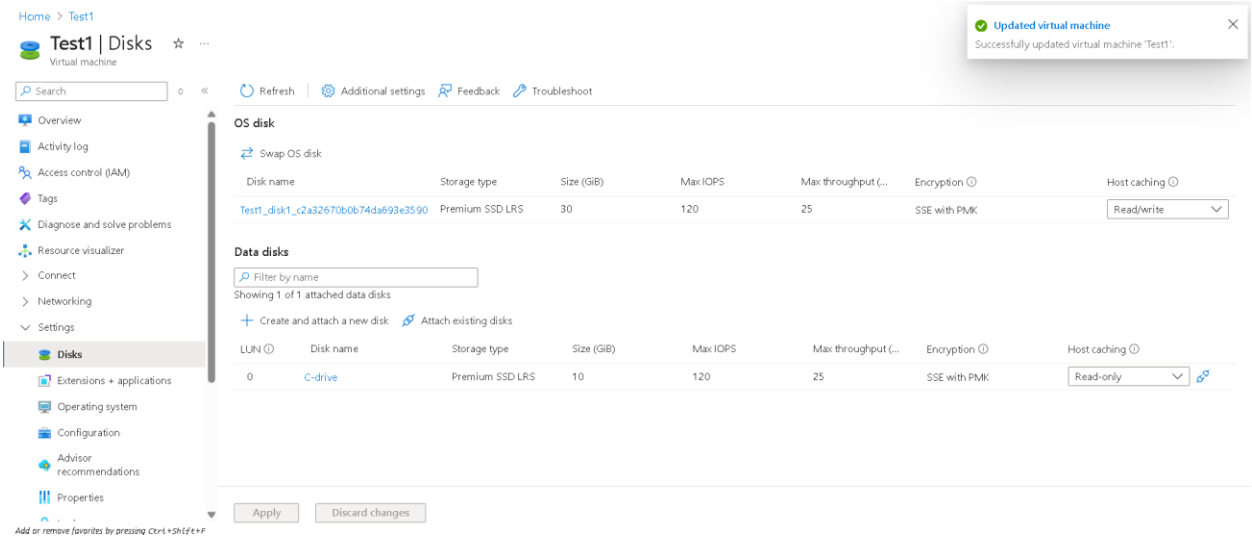




5. Connect via SSH

- **Action**: Connected to the VM using an SSH client Linux terminal.
- **Command Used**:
  `ssh -i path_to_private_key.pem azureuser@<public-ip-address>`



6. Create & Attach Additional Block Storage

- **Action**: Provisioned and attached an additional managed disk.
- **Steps Taken**:
  - Navigated to **Disks > Add Data Disk** under the VM settings.
  - Created a **Standard HDD** of 10 GB, attached it to the VM.